

# Predictors of COVID-19 Protective Behavior Before and After Mask Mandates: A Cross-National Analysis

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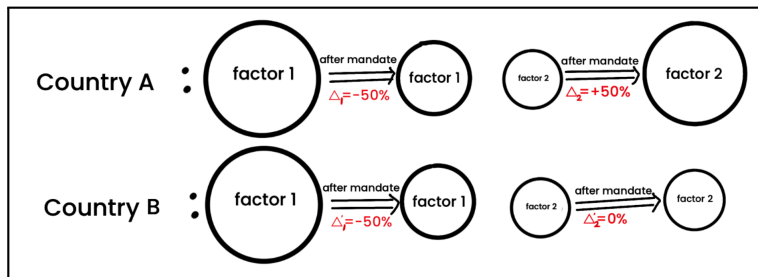
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## 1. Project background

- During **COVID-19**, protective behaviours such as **mask wearing**, social distancing, hand hygiene, and self-isolation were important for reducing transmission.
- Compliance was shaped not only by policy rules, but also by risk perception, trust, demographic characteristics, wellbeing, and broader social context.
- These predictors may operate **differently across countries** because policy environments, institutions, culture, and public expectations vary (Ryan et al., 2025).

## 2. Research questions

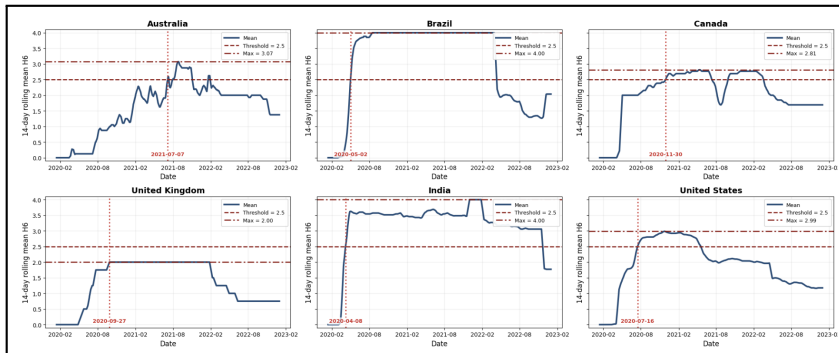
- Within each country, do the key predictors of compliance with COVID-19 protective behaviours change before and after mask mandates?
- Are these changes consistent across countries, or do they vary?
- Is there a set of common cross-national predictors, alongside country-specific predictors?



## 1. Behaviour and policy datasets

- **YouGov dataset:** repeated survey responses about people's COVID-19 behaviours, attitudes, and background characteristics (Imperial College London and YouGov, 2022).
  - Behaviour: whether people wore masks, kept physical distance, washed hands...
  - Predictor: age group, gender, confidence in health services...
- **OxCGRT dataset:** This dataset contains the strictness with which the government has enforced its mask-wearing policy (Oxford COVID-19 Government Response Tracker, 2023).
- The two sources are linked by **country, region and survey date**.

## 2. Policy timing figure



This figure show how mask policy intensity changed over time, and how the mandate threshold was used to define the before and after mandate periods.

## 3. Country coverage

The current cleaned cross-national sample includes six countries:

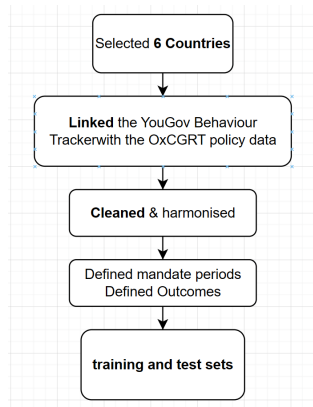
| Country code   | Country name     |
|----------------|------------------|
| AUS            | Australia        |
| BRA            | Brazil           |
| CAN            | Canada           |
| <del>CHN</del> | <del>China</del> |
| GBR            | United Kingdom   |
| IND            | India            |
| USA            | United States    |

China was excluded from the current analysis because several key variables needed for the harmonised workflow were not available.

# What Have I Done So Far?

## 1. Data pre-processing

- For the YouGov and OxCGRT files, various formatting issues have been resolved.
- **Mapped survey regions to policy regions:** Convert all place names in both datasets to lower-case English letters, then match them one-to-one.
- **Defined before/after mandate periods.**
- **Defined two outcomes:** Face-mask-wearing and Overall-protective-behaviour.
- **Cleaned the data:** Removing columns and samples with too many missing values, encoding the data, making column names more readable, and splits.



## 2. About Model

### **Model 1: Random Forest**

This model builds many decision trees and lets them vote for the final prediction. Capable of handling non-linearity.

### **Model 2: XGBoost**

This model also uses decision trees, but it builds them step by step. Each new tree tries to correct the mistakes made by the previous trees. Capable of handling non-linearity.

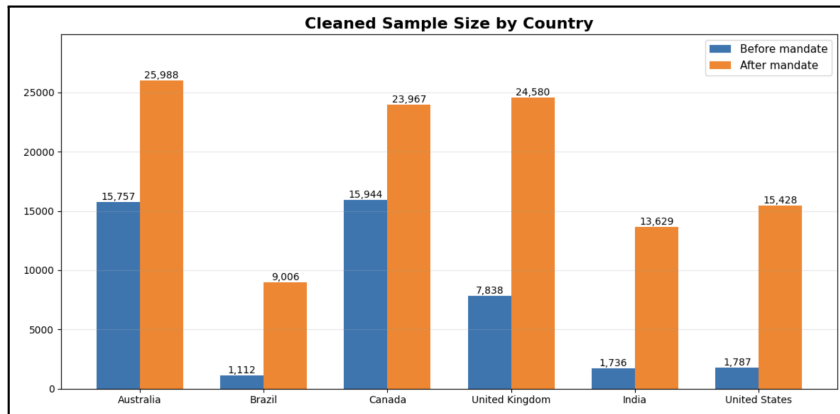
### **Metric: AUC**

AUC means Area Under the ROC Curve; selected because it can measure model separation across all classification thresholds. AUC ranges from 0 to 1. A higher AUC indicates better separation between compliant and non-compliant respondents.



## 1. Cleaned sample size

The preprocessing pipeline produced cleaned before- and after-mandate datasets for all six countries.



## 2. Train-test split

The next step was to prepare data for modelling.

- I split the data separately for each country, policy period, and outcome.
- I used an 80/20 train-test split.
- Stratification (based on the outcome class, maintain a similar proportion of categories in the train and test sets) was used where possible to preserve the class balance of the outcome.

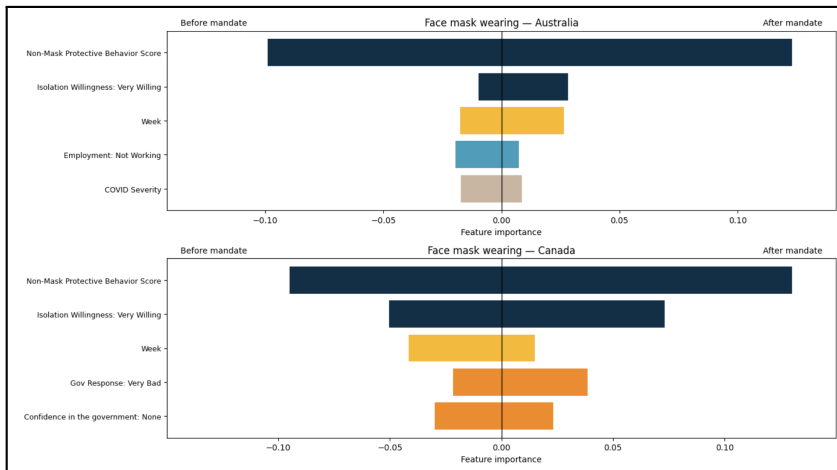
| Dimension             | Values                                  | Number |
|-----------------------|-----------------------------------------|--------|
| Countries             | AUS, BRA, CAN, GBR, IND, USA            | 6      |
| Policy periods        | Before mandate, after mandate           | 2      |
| Outcomes              | Face mask, overall protective behaviour | 2      |
| Total modelling tasks | $6 \times 2 \times 2$                   | 24     |

### 3. Model Performance (AUC)

| Country   | Outcome                             | RF           | XGboost      | Country | Outcome                             | RF           | XGboost      |
|-----------|-------------------------------------|--------------|--------------|---------|-------------------------------------|--------------|--------------|
| Australia | Face mask wearing Before mandate    | 0.844        | <b>0.854</b> | UK      | Face mask wearing Before mandate    | 0.755        | <b>0.760</b> |
|           | Face mask wearing After mandate     | 0.909        | <b>0.912</b> |         | Face mask wearing After mandate     | 0.695        | <b>0.705</b> |
|           | Protective behaviour Before mandate | 0.767        | <b>0.772</b> |         | Protective behaviour Before mandate | 0.796        | <b>0.797</b> |
|           | Protective behaviour After mandate  | 0.826        | <b>0.831</b> |         | Protective behaviour After mandate  | 0.846        | <b>0.857</b> |
| Brazil    | Face mask wearing Before mandate    | 0.783        | <b>0.785</b> | India   | Face mask wearing Before mandate    | 0.757        | <b>0.764</b> |
|           | Face mask wearing After mandate     | 0.652        | <b>0.653</b> |         | Face mask wearing After mandate     | <b>0.856</b> | 0.851        |
|           | Protective behaviour Before mandate | <b>0.713</b> | 0.705        |         | Protective behaviour Before mandate | <b>0.823</b> | 0.801        |
|           | Protective behaviour After mandate  | <b>0.829</b> | 0.827        |         | Protective behaviour After mandate  | 0.827        | <b>0.831</b> |
| Canada    | Face mask wearing Before mandate    | <b>0.854</b> | <b>0.854</b> | US      | Face mask wearing Before mandate    | 0.846        | <b>0.853</b> |
|           | Face mask wearing After mandate     | 0.813        | <b>0.816</b> |         | Face mask wearing After mandate     | 0.855        | <b>0.862</b> |
|           | Protective behaviour Before mandate | 0.836        | <b>0.840</b> |         | Protective behaviour Before mandate | 0.797        | <b>0.802</b> |
|           | Protective behaviour After mandate  | 0.833        | <b>0.834</b> |         | Protective behaviour After mandate  | 0.857        | <b>0.862</b> |

## 4. Feature importance

Feature importance indicates how much each predictor contributes to the model's prediction. A higher importance score means that the factor plays a stronger role (Calculated after model training based on the contribution of each variable to the predicted results).



## Summary of main findings

- **XGBoost** achieved the **best** or joint-best AUC in most modelling tasks, while Random Forest provided a useful comparison model with similar performance in some cases.
- The before/after comparison suggests that **the predictors of compliance are not fixed**. Their importance changes after mask mandates, and the direction of change differs across countries.
- **Some predictors appear repeatedly across countries**, such as non-mask protective behaviour, willingness to self-isolate, survey timing, and confidence-related variables. Other predictors are more country-specific.

## All data and code

All code can be found in my GitHub: <https://github.com/ZixaunZhao/COVID-19-Protective-Behavior-Cross-National-Analysis.git>

- [1] Imperial College London and YouGov. COVID-19 behaviour tracker data hub, 2022. URL: <https://github.com/YouGov-Data/covid-19-tracker/tree/master/data>. Accessed 2026.
- [2] Oxford COVID-19 Government Response Tracker. COVID-19 policy tracker data, 2023. URL: <https://github.com/OxCGRT/covid-policy-tracker/tree/master/data>. Accessed 2026.
- [3] Ryan, M., Ye, J., Sexton, J., Hickson, R. I., and Brindal, E. Face mask mandates alter major determinants of adherence to protective health behaviours in Australia. Royal Society Open Science, 12:241941, 2025. DOI: 10.1098/rsos.241941.